

Governance-Aware Hybrid Decision Fusion for Data Quality Anomaly Detection in Cloud Lakehouses: A Multi-Domain Evaluation

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Abstract

Cloud lakehouses increasingly combine deterministic validation rules with unsupervised anomaly detectors, yet their outputs are often evaluated separately and converted into operational actions through opaque thresholds. This study presents a governance-aware, policy-based fusion framework that combines rule risk, normalized AI consensus, detector voting, and evidence uncertainty into auditable record-level risk and complementary trust indices, then routes records to accept, repair, quarantine, or escalate actions. Unlike a learned stacking model, the fusion function remains inspectable and versionable to support replay, policy control, and accountable intervention. The framework was evaluated in 63 controlled experiments spanning finance, healthcare, and retail data. Seven anomaly classes were injected at 1%, 5%, and 10% rates. Baselines included rule-only validation, Isolation Forest, Local Outlier Factor, and AI consensus. A 3 x 3 nested experiment-level cross-validation procedure selected the same constrained policy in all outer folds: weights of 0.20 for rule risk, 0.55 for AI consensus, 0.20 for detector vote, and 0.05 for uncertainty, with a threshold of 0.45. AI consensus achieved the highest numerical macro-F1 (0.306), while the hybrid achieved 0.281; the difference was not significant after Holm correction (mean -0.025, 95% bootstrap confidence interval [-0.075, 0.019], adjusted $p = 0.482$). The hybrid reduced false-positive rate by 0.057 relative to the initial hybrid policy (adjusted $p < 0.001$) and exceeded rule-only F1 by 0.242 (adjusted $p < 0.001$). The results do not establish predictive superiority or formal equivalence, but show stable, domain-dependent evidence governance with explicit decision provenance and controlled false-positive behavior. Evaluation against modern deep-learning anomaly detectors was intentionally left outside the scope of this study and is identified as future work.

Keywords: data quality; anomaly detection; knowledge engineering; decision fusion; cloud lakehouse; data governance; auditable AI

1. Introduction

Data quality has become a continuous operational concern in analytical platforms rather than a one-time preprocessing task. Modern cloud lakehouses combine low-cost object storage, transactional table formats, scalable computation, and direct support for business intelligence and machine-learning workloads [1–3]. The same convergence increases the consequences of incomplete, inconsistent, duplicated, drifting, or contextually implausible records because defects can propagate from ingestion into dashboards, feature pipelines, automated decisions, and model retraining.

Deterministic validation rules remain essential because they encode explicit domain knowledge and provide clear reasons for rejection. They are particularly effective for schema violations, missing identifiers, impossible ranges, uniqueness constraints, and known cross-field dependencies. However, rule systems are bounded by what has been specified in advance. They can miss multivariate, contextual, distributional, or previously unseen anomalies, and they often become difficult to maintain when data products evolve across domains [4–6].

Unsupervised anomaly detectors address a different part of the problem. Isolation Forest and Local Outlier Factor, for example, identify globally isolated observations and locally sparse neighborhoods without requiring a fully labeled anomaly corpus [7,8]. Their scores can reveal unusual records that pass deterministic constraints. At the same time, score thresholds, detector disagreement, domain shift, and class imbalance can produce unstable operational behavior. A detector that improves recall can also create an impractical review queue, while a conservative threshold can conceal defects behind a low false-positive rate [9,10].

The systems challenge is therefore broader than selecting the detector with the highest F1 score. A governed data platform must combine heterogeneous evidence, preserve the reason for each decision, control the false-positive burden, and route records to actions that are consistent with policy. These requirements align with data-governance principles concerning accountability, decision rights, metadata, and stewardship [11,12]. A limited evidence-first analogy can also be drawn from Zero Trust [13], but this study does not implement the NIST security architecture or present the framework as a Zero Trust system.

Existing research addresses data-quality measurement, anomaly detection, explainability, and governance, but the evidence streams are commonly studied in isolation. Benchmarks typically compare models and scores, while operational systems often rely on static rules or proprietary threshold logic. Explainable anomaly-detection research improves local interpretation, yet explanation alone does not specify whether a record should be accepted, repaired, quarantined, or escalated [14–17]. Conversely, governance frameworks establish responsibilities but rarely quantify the predictive and operational trade-offs of combining rule and AI evidence.

This study investigates a governance-aware hybrid decision framework designed for record-level data-quality control in cloud lakehouses. The framework combines a deterministic rule-risk score, normalized AI consensus, detector voting, and evidence uncertainty. It produces a bounded risk score, its complementary trust score, and one of four governed actions. The design makes the decision function explicit and versionable so that a reviewer can reconstruct why a record was routed to a particular state.

The evaluation deliberately tests a stronger claim than simple feasibility. Sixty-three controlled experiments were formed from three public tabular datasets, seven anomaly classes, and three injection rates. Rule-only validation, Isolation Forest, Local Outlier Factor, AI consensus, and three generations of hybrid policy were compared. Final policy selection used nested experiment-level cross-validation with a false-positive constraint, followed by paired bootstrap intervals, Wilcoxon signed-rank tests, Friedman omnibus tests, and Holm correction.

The study addresses four research questions: (RQ1) How does hybrid evidence fusion compare with rule-only and AI-only methods across domains and anomaly classes? (RQ2) Can constrained calibration reduce the false-positive burden of an initial hybrid policy without producing an unsupported claim of predictive superiority? (RQ3) Is the selected policy stable across independent experiment-level folds? (RQ4) What operational review burden results from the competing methods?

The contributions are fourfold. First, the paper specifies an auditable, policy-based evidence-fusion and routing mechanism whose coefficients, threshold, evidence terms, and actions can be inspected and versioned; it is not presented as a learned meta-model or as another anomaly detector. Second, it provides a multi-domain controlled benchmark with seven defect classes and explicit ground truth. Third, it uses nested experiment-level calibration to avoid record-level leakage and post-hoc threshold selection. Fourth, it reports a statistically supported negative result: the governed hybrid is not universally more accurate than AI consensus, although it is stable, significantly improves on rule-only validation, and significantly reduces false positives relative to the initial hybrid policy.

2. Background and related work

2.1 Data quality and deterministic validation

Wang and Strong conceptualized data quality as fitness for use and emphasized that accuracy alone is insufficient [4]. Subsequent work organized quality concerns into dimensions such as completeness, consistency,

timeliness, and accuracy, while survey research documented the diversity of assessment and improvement methods [5]. These foundations motivate rule engines that translate known expectations into executable checks.

Rule-based validation offers high interpretability: a failed age range, missing identifier, nonpositive price, or duplicate key can be explained directly. It also supports reproducibility because the same versioned rule applied to the same record yields the same outcome. The limitation is coverage. Rules require prior knowledge of defect patterns, and their interactions become harder to manage as schemas and business semantics change. A rule-only system may therefore have a low alert rate while still missing most injected anomalies.

Recent empirical research confirms that data defects affect downstream machine-learning behavior, but effects vary by quality dimension, algorithm, and whether contamination occurs in training or test data [18]. This variation supports evaluating data-quality controls across multiple defect types and domains rather than assuming a single detector or quality dimension generalizes.

2.2 Unsupervised anomaly detection for tabular data

Isolation Forest isolates observations through random recursive partitioning; anomalies tend to require shorter paths [7]. Local Outlier Factor estimates how isolated an observation is relative to its local neighborhood [8]. The two methods provide complementary global and local views and are commonly used when labels are limited.

Anomaly-detection surveys distinguish point, contextual, and collective anomalies and note that performance depends on the application, representation, and evaluation protocol [9]. Tabular quality defects may appear as impossible values, rare combinations, contextual inconsistencies, or distributional shifts. A detector that performs well on global outliers may not identify a semantically invalid but statistically common record, which motivates fusing learned evidence with domain rules.

Explainable anomaly-detection work has proposed feature- or cell-level explanations for tabular anomalies, Shapley-based identification of quality issues, and benchmarks for explanation quality [14–17]. These methods improve diagnostic value, but the governance problem remains: explanations and scores must still be converted into policy-controlled actions, recorded with provenance, and evaluated against operational costs.

2.3 Governance, auditability, and cloud lakehouses

Data governance defines decision rights, accountabilities, and controls for data assets [11,12]. In a lakehouse, those controls must operate across ingestion, curated tables, feature construction, and analytical consumption. Transactional table formats and scalable engines make it possible to preserve versions and lineage, but they do not determine whether a suspect record should be repaired or quarantined [1–3].

NIST Zero Trust Architecture is a security framework concerned with explicit authorization and continuous evaluation before access to resources [13]. It is not implemented in this study. The narrower principle used here is evidentiary: a record is not accepted merely because it passed ingestion or arrived from an approved source. Acceptance depends on current, versioned evidence, and the evidence, policy, and outcome remain auditable. This analogy motivates the absence of implicit data trust without extending the contribution into identity, network, or access-control architecture.

FAIR and dataset-documentation principles further emphasize provenance, reuse, and transparent descriptions of data resources [19,20]. Data cascades and hidden technical debt show how upstream data problems can create downstream failures that are difficult to diagnose [21,22]. These concerns motivate a decision layer that stores not only anomaly predictions but also the evidence and reason for intervention.

2.4 Learned and probabilistic evidence fusion

Model outputs can be combined through stacking, Bayesian model averaging, and performance-optimized ensemble selection. Stacked generalization trains a second-level learner on out-of-sample predictions from the base models [31]. Bayesian model averaging weights predictions using posterior model probabilities and explicitly represents uncertainty about model choice [32]. Ensemble selection searches a model library to optimize a chosen predictive

metric [33]. These approaches can learn data-dependent weights or nonlinear interactions and may outperform fixed aggregation when sufficient representative meta-training data are available.

A learned combiner also creates a second model lifecycle: unbiased meta-features must be generated, calibration and leakage must be controlled, retraining triggers must be defined, and the resulting decisions require their own explanations and monitoring. This study therefore evaluates an explicit policy function whose terms and constraints can be reviewed before deployment and replayed afterward. The choice isolates the governance properties of evidence fusion; it is not a claim that a fixed linear policy is predictively optimal. Learned and probabilistic combiners remain important comparators for subsequent work.

2.5 Research gap

The literature provides mature components - quality dimensions, rule checks, unsupervised detectors, learned ensemble methods, explanation techniques, lakehouse storage, and governance principles - but lacks a consistently evaluated mechanism that converts heterogeneous evidence into auditable record-level actions while quantifying both predictive and operational trade-offs. It also seldom distinguishes a predictive ensemble, whose primary objective is accuracy, from a governance policy, whose objective includes traceability, review burden, and enforceable action semantics. The contribution of this study is therefore systems-oriented: it formalizes policy-based evidence fusion and routing, evaluates calibration without record-level leakage, and establishes the boundary between predictive accuracy and governance value.

3. Governance-aware hybrid decision framework

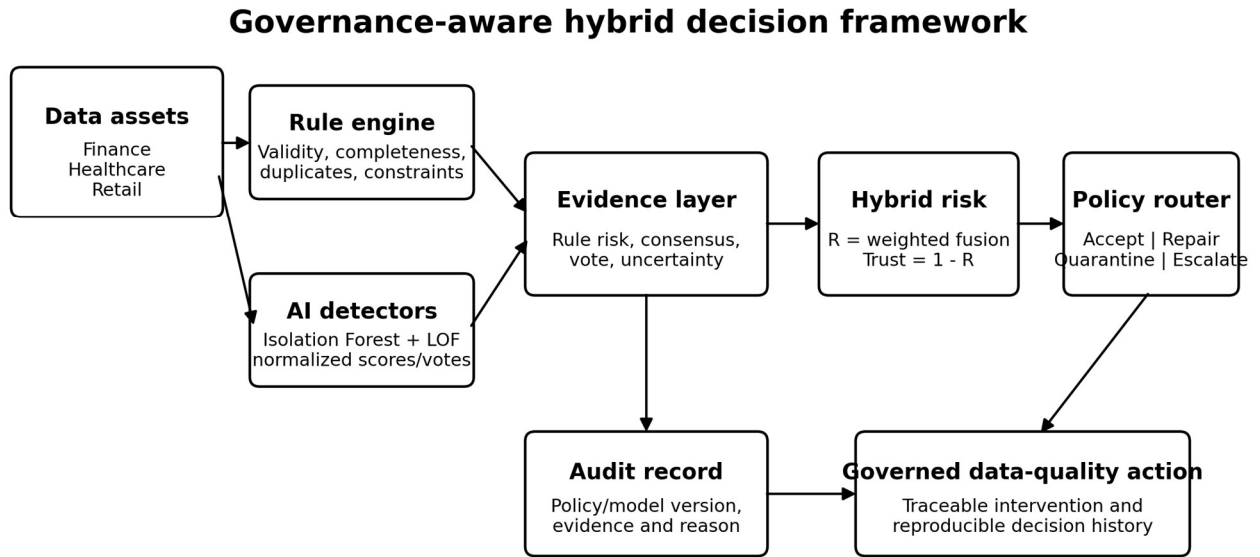


Figure 1. Governance-aware evidence fusion and policy routing. The framework combines deterministic and learned evidence, generates bounded risk and trust scores, and records the decision context required for audit and reproduction.

3.1 Evidence representation

For record i , the framework constructs four normalized evidence components in $[0,1]$. The rule-risk score r_i aggregates deterministic violations and their severity. The AI consensus score c_i combines normalized Isolation Forest and Local Outlier Factor scores. The detector-vote score v_i is the fraction of binary detector predictions that identify the record as anomalous. The uncertainty score u_i summarizes incomplete evidence, score disagreement, and rule–AI conflict.

The components are deliberately separated. Rule risk represents explicit domain knowledge; consensus preserves continuous anomaly strength; voting captures discrete detector agreement; and uncertainty prevents a confident operational action when the supporting evidence is incomplete or contradictory. Each component, its source model or rule version, and the resulting action are retained in an audit record.

3.2 Risk, complementary trust index, and calibrated policy

The final constrained configuration, selected independently in all three outer folds, computes the hybrid risk score as:

$$R(i) = 0.20 r(i) + 0.55 c(i) + 0.20 v(i) + 0.05 u(i) \quad (1)$$

The corresponding trust score is:

$$T(i) = 1 - R(i) \quad (2)$$

A record is predicted as anomalous when $R(i) \geq 0.45$. The weights sum to one and the threshold was selected only within inner validation folds under a false-positive-rate constraint. The same candidate (W08) and threshold were selected in all outer folds, providing evidence of configuration stability. $T(i)$ is a deterministic complement used for policy reporting; it is not a calibrated probability that a record is trustworthy.

3.3 Design rationale for policy-based fusion

The framework uses an explicit weighted policy rather than a learned meta-classifier. This is a design choice, not a claim that linear fusion maximizes F1. In operational data-quality systems, decisions may need to be reviewed before deployment, reconstructed after an incident, compared across policy versions, and approved by data owners or stewards. Fixed coefficients and a threshold can be represented as configuration, inspected independently of code, and replayed without retraining when only the governance policy changes.

A learned combiner - such as logistic stacking, a boosted meta-model, or Bayesian model averaging - could model interactions among evidence sources and potentially improve predictive performance. It would also introduce meta-training requirements, additional calibration and leakage risks, retraining and drift-monitoring obligations, and a separate explanation boundary. The present design treats predictive optimization as a constrained objective and asks whether an auditable policy can remain stable while controlling false positives. Learned combiners are not assumed to be inferior and are identified as a necessary future comparison.

3.4 Governed decision actions

The binary prediction is converted into a governed action. ACCEPT is assigned when risk is below the calibrated threshold. REPAIR is used when a deterministic rule fails but AI evidence is not sufficiently strong to justify isolation, indicating a known and potentially correctable defect. QUARANTINE is used for critical rule severity or strong AI evidence. ESCALATE is used when evidence is incomplete or when high detector disagreement coincides with rule-AI conflict.

The action taxonomy separates predictive detection from operational handling. Two records can have similar risk but require different responses because one has a known correctable violation and the other has conflicting evidence. The policy router therefore acts as a knowledge-engineering layer: it encodes how evidence should be interpreted and records the decision rationale rather than hiding it inside a model threshold.

3.5 Auditability and evidence-based trust

Every decision record contains the experiment identifier, data domain, record identifier, rule outcome, detector scores and votes, evidence completeness, uncertainty, risk, complementary trust index, threshold, policy version, and action. This structure supports replay, comparison across policy versions, and traceability from an intervention to the evidence that justified it. The approach does not implement NIST SP 800-207; it borrows only a narrow evidence-first principle. Records are not accepted solely because they originate from an approved pipeline, and acceptance requires explicit, versioned evidence. Identity, network, and access-control mechanisms remain outside the scope of the framework.

4. Experimental design

4.1 Public datasets

Three public tabular datasets were selected to represent distinct operational domains and schema characteristics. The Bank Marketing dataset provides mixed demographic, campaign, and financial attributes [23]. The Diabetes 130-US Hospitals dataset contains encounter-level clinical and utilization variables [24]. Online Retail II contains invoice-level transactions with product, quantity, price, customer, and time attributes [25]. All datasets are de-identified public research resources; no new human-subject data were collected.

Table 1. Dataset coverage and out-of-fold evaluation observations.

Dataset	Domain	Original public records	Experiments	OOF record-experiment observations
Bank Marketing	Finance	45,211	21	420,000
Diabetes 130-US Hospitals	Healthcare	101,766	21	141,855
Online Retail II	Retail	1,067,371	21	126,000
Total	Three domains	—	63	687,855

The out-of-fold count is the number of record–experiment observations scored during the 63 experiments, not the sum of unique source rows. Repeated controlled experiments intentionally evaluate different anomaly classes and rates over domain-specific samples.

4.2 Deterministic rule engine

The rule engine encoded domain-relevant completeness, validity, and duplication checks. Finance checks included missing age, age outside 18–100, negative balance, negative campaign count, and duplicate combinations of demographic and campaign fields. Healthcare checks included missing encounter or patient identifiers, hospital stay outside 1–14 days, negative inpatient counts, and duplicate encounters. Retail checks included missing invoice or stock code, zero quantity, missing or nonpositive price, and duplicate invoice-line combinations. Rule failures generated a binary prediction, severity, and bounded rule-risk score.

4.3 Controlled anomaly taxonomy

Seven anomaly classes were injected at 1%, 5%, and 10% rates, creating 3 datasets $\times$ 7 anomaly classes $\times$ 3 rates = 63 experiments. Each injected row received an explicit ground-truth label.

Table 2. Controlled anomaly classes and representative transformations.

Code	Anomaly class	Representative transformation
A1	Missing value	Nullify a key field such as balance, patient identifier, or stock code.
A2	Duplicate record	Duplicate an existing record or domain key.
A3	Domain-rule violation	Set impossible values, e.g., age = −10, stay = 50, or price = −100.
A4	Statistical outlier	Multiply a numeric value such as balance, inpatient count, or quantity by 8.
A5	Contextual anomaly	Create an extreme value that is implausible only in a selected context.
A6	Distribution drift	Shift and scale a numeric distribution for selected records.

Code	Anomaly class	Representative transformation
A7	Cross-attribute inconsistency	Create semantically inconsistent combinations across two attributes.

4.4 AI baselines and consensus

Numeric variables were median-imputed and robustly scaled. Categorical variables were most-frequent imputed and one-hot encoded, with cardinality controls to limit memory growth. Isolation Forest used 200 trees with a fixed random seed. Local Outlier Factor used novelty detection with an adaptive neighborhood size. Models were fitted independently within each experiment.

The AI-consensus baseline combined normalized detector scores and binary votes while preserving individual detector outputs for analysis. This baseline was intentionally strong: it represents a practical detector ensemble without deterministic rule-risk or governance uncertainty terms. The complete comparison also included each detector separately and the rule-only engine.

4.5 Hybrid policy generations and nested validation

Hybrid V1 was a manually specified prototype used to verify the evidence schema and action routing. Its relatively aggressive threshold increased recall but also produced a high false-positive rate. Hybrid V2 searched weight and threshold combinations on a calibration split and improved holdout F1, but the held-out false-positive rate remained high. Because that holdout had then been inspected, it was not used for further tuning.

Hybrid V3 used 3-fold outer and 3-fold inner stratified cross-validation at the experiment level. Experiments—not individual records—were the splitting unit, preventing rows from the same experimental condition from appearing in both policy selection and evaluation. In each outer fold, inner validation selected the candidate with the highest macro-F1 subject to a mean FPR no more than 0.02 above the AI-consensus validation FPR. If no candidate satisfied the constraint, the minimum-violation candidate would have been selected; this fallback was never required.

The candidate grid varied the weights assigned to rule risk, AI consensus, detector vote, and uncertainty, while thresholds ranged from 0.25 to 0.65. The independently selected W08 configuration was identical in all outer folds: 0.20, 0.55, 0.20, and 0.05, respectively, with a threshold of 0.45.

4.6 Evaluation metrics and statistical analysis

For each experiment and method, true positives, true negatives, false positives, and false negatives were used to compute precision, recall, F1, FPR, FNR, specificity, balanced accuracy, and area under the precision–recall curve. Macro averages give equal weight to each experiment and reduce domination by large datasets.

Paired experiment-level differences were evaluated using 10,000 bootstrap resamples and two-sided Wilcoxon signed-rank tests. Holm correction controlled family-wise error across pairwise tests. Friedman tests assessed omnibus differences across four principal methods. Statistical significance was set at 0.05 after correction. Operational burden was quantified by flagged-record rate, anomaly-capture rate, false reviews per true positive, and missed anomalies.

5. Results

5.1 Overall detection performance

AI consensus achieved the highest macro-F1 (0.306), followed by Hybrid V1 (0.293), Local Outlier Factor (0.291), Hybrid V3 (0.281), Isolation Forest (0.112), and rule-only validation (0.039). Hybrid V3 achieved the highest precision among the three nested-validation methods (0.426), but its constrained threshold reduced recall to

0.416. Local Outlier Factor produced the largest PR-AUC, illustrating that ranking performance and thresholded F1 need not select the same method.

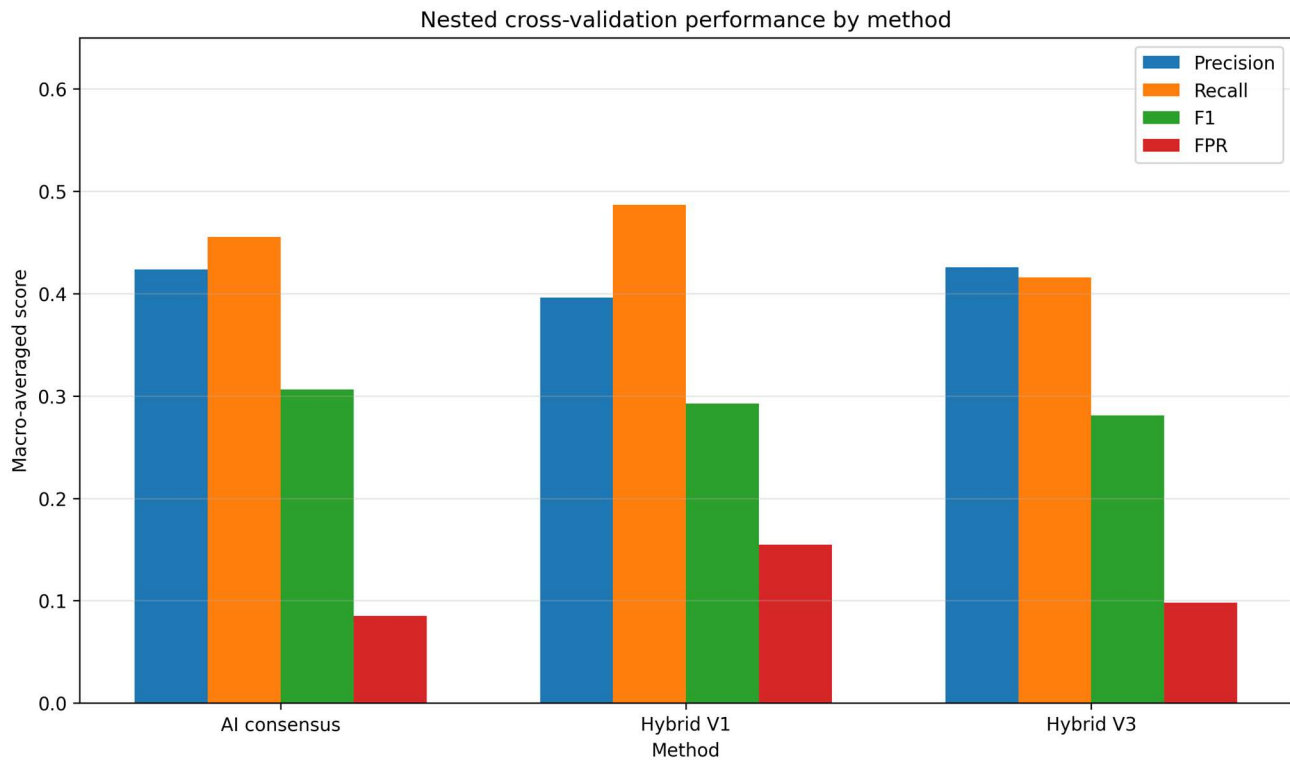


Figure 2. Nested experiment-level macro precision, recall, F1, and false-positive rate for AI consensus, Hybrid V1, and Hybrid V3.

Table 3. Macro performance across 63 experiments. Hybrid V3 values are out-of-fold predictions from nested validation.

Method	Precision	Recall	F1	FPR	FNR	Bal. acc.	PR-AUC
Rule-only	0.050	0.043	0.039	0.050	0.957	0.385	0.104
Isolation Forest	0.358	0.153	0.112	0.045	0.847	0.443	0.190
Local Outlier Factor	0.477	0.375	0.291	0.048	0.625	0.552	0.307
AI consensus	0.424	0.455	0.306	0.085	0.545	0.574	0.259
Hybrid V1	0.396	0.487	0.293	0.155	0.513	0.555	0.289
Hybrid V3	0.426	0.416	0.281	0.098	0.584	0.548	0.290

5.2 Paired statistical comparisons

The numerical F1 difference between Hybrid V3 and AI consensus was -0.0254 . Its 95% paired bootstrap confidence interval ranged from -0.0748 to 0.0187 , and the Holm-adjusted Wilcoxon p-value was 0.482 . The evidence therefore does not establish a statistically significant F1 difference. Precision was also statistically indistinguishable (difference 0.0020 ; adjusted $p = 0.581$), as was recall (difference -0.0393 ; adjusted $p = 1.000$).

Hybrid V3 had a significantly higher FPR than AI consensus by 0.0133 (95% CI $[0.0081, 0.0186]$, adjusted $p < 0.001$). However, it reduced FPR by 0.0566 relative to Hybrid V1 (95% CI $[-0.0656, -0.0469]$, adjusted $p < 0.001$) and increased precision by 0.0295 (adjusted $p < 0.001$). Hybrid V3 exceeded rule-only F1 by 0.2419 (95% CI $[0.1924, 0.2951]$, adjusted $p < 0.001$). Friedman tests were significant for all evaluated metrics, confirming that the methods did not share the same performance distribution.

Table 4. Key paired experiment-level comparisons. Differences are method A minus method B.

Metric	Comparison	Mean diff.	95% bootstrap CI	Holm-adjusted p	Result
F1	Hybrid V3 – AI consensus	-0.0254	$[-0.0748, 0.0187]$	0.482	Not significant
Precision	Hybrid V3 – AI consensus	0.0020	$[-0.0074, 0.0114]$	0.581	Not significant
Recall	Hybrid V3 – AI consensus	-0.0393	$[-0.1009, 0.0155]$	1.000	Not significant
FPR	Hybrid V3 – AI consensus	0.0133	$[0.0081, 0.0186]$	<0.001	Significant
FPR	Hybrid V3 – Hybrid V1	-0.0566	$[-0.0656, -0.0469]$	<0.001	Significant
F1	Hybrid V3 – Rule-only	0.2419	$[0.1924, 0.2951]$	<0.001	Significant

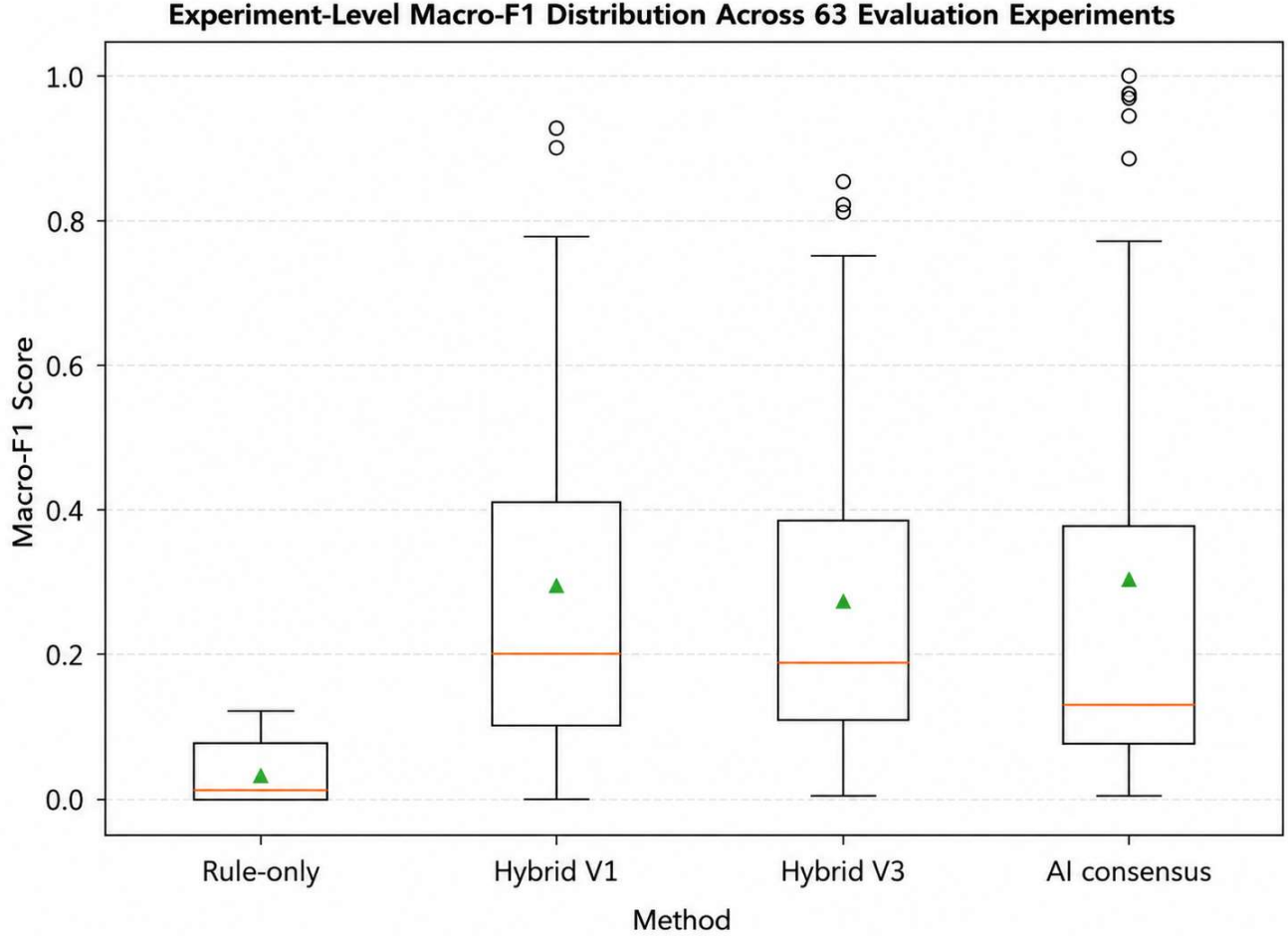


Figure 3 presents the distribution of experiment-level macro-F1 scores across the 63 evaluation experiments. Although AI consensus achieved the highest median and mean F1, substantial overlap exists between the Hybrid V3 and AI consensus distributions, visually supporting the paired statistical tests reported in Table 4. In contrast, the rule-only baseline consistently produced the lowest performance across experiments.

5.3 Domain-specific behavior

Performance varied materially by domain. Hybrid V3 improved F1 over AI consensus in finance by 0.0205 and in retail by 0.0121, but underperformed by 0.1087 in healthcare. The healthcare result combined very high precision (0.863) with low recall (0.259), indicating that the globally constrained threshold was conservative for this domain. Retail showed the opposite profile: high recall (0.665) and low precision (0.162). These findings reject an assumption that one global fusion policy produces uniform benefits.

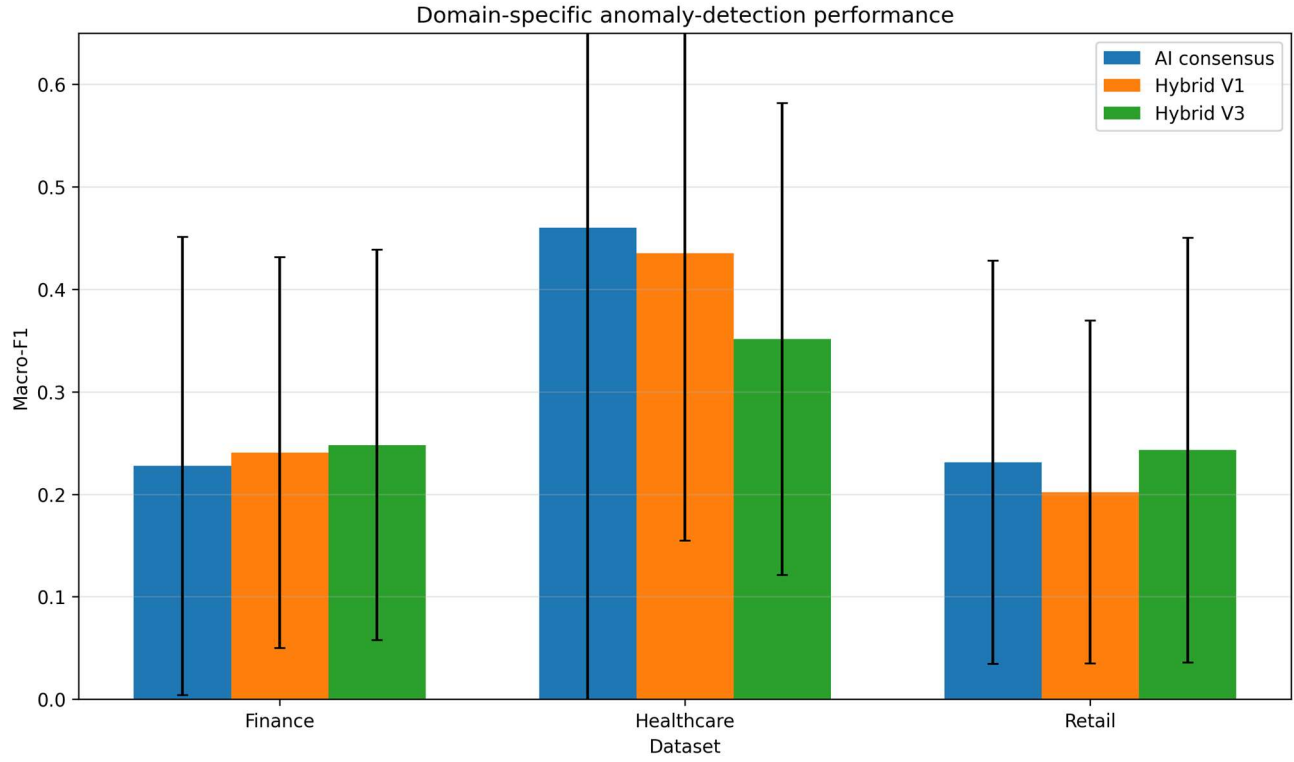


Figure 4. Domain-specific macro-F1 for AI consensus, Hybrid V1, and Hybrid V3.

Table 5. Hybrid V3 out-of-fold performance by domain.

Dataset	Experiments	OOF observations	Precision	Recall	F1	FPR	FNR	PR-AUC
Finance	21	420,000	0.252	0.324	0.248	0.108	0.676	0.248
Healthcare	21	141,855	0.863	0.259	0.352	0.018	0.741	0.580
Retail	21	126,000	0.162	0.665	0.243	0.168	0.335	0.236

5.4 Policy stability and experiment-level variation

All three outer folds independently selected W08 with threshold 0.45 and satisfied the FPR constraint. Inner-validation F1 values were 0.276, 0.289, and 0.278, while inner FPR values were 0.100, 0.098, and 0.097. This repeated selection is stronger evidence of policy stability than choosing one configuration after inspecting all experiments.

Hybrid V3 achieved higher experiment-level F1 than AI consensus in 41 of 63 experiments and lower F1 in 22, despite a negative mean difference. The result reflects unequal effect magnitudes: relatively small gains in finance and retail were outweighed by larger healthcare losses. Against Hybrid V1, Hybrid V3 won 32 experiments and lost 31 on F1, but had lower FPR in 49 and equal FPR in 14, with no experiment having higher FPR.

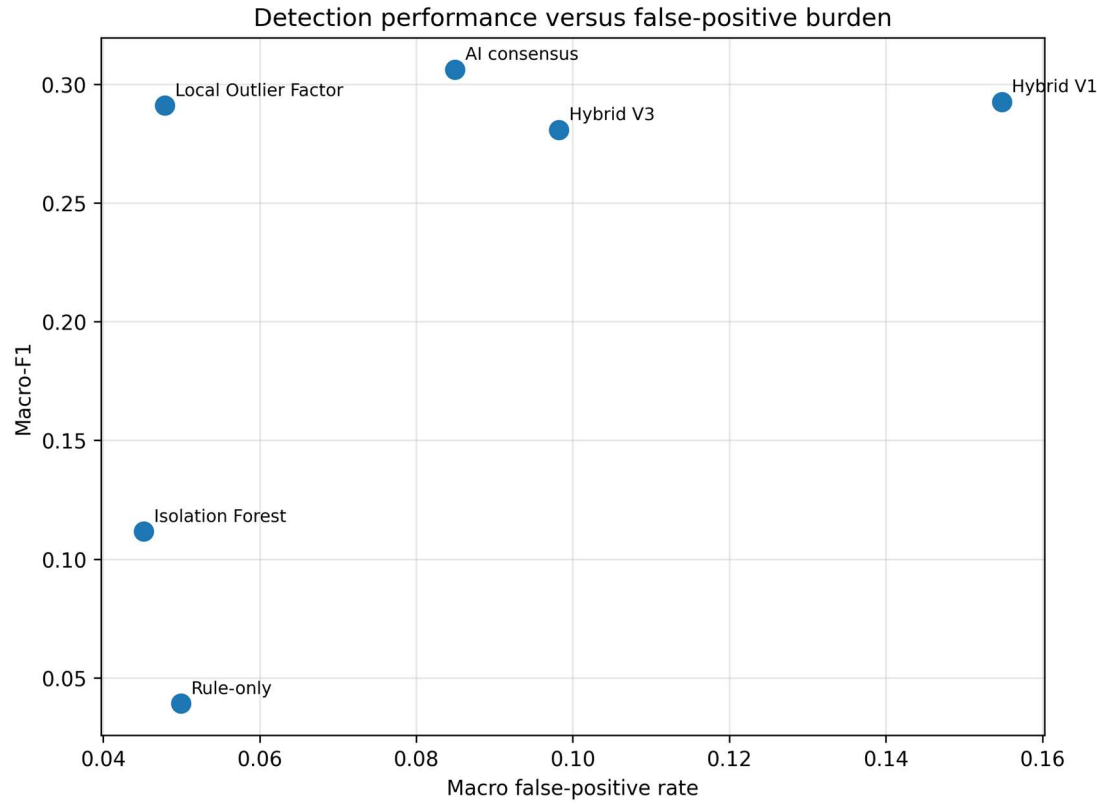


Figure 5. Macro-F1 versus macro false-positive rate. The upper-left region represents higher F1 with lower false-positive burden.

5.5 Operational intervention burden

Operational metrics clarify the cost of threshold choices. Across 687,855 out-of-fold observations containing 171,269 injected anomalies, AI consensus flagged 17.3% of records and captured 40.1% of anomalies, with 0.729 false reviews per true positive. Hybrid V3 flagged fewer records (15.1%) but captured only 24.3% of anomalies and generated 1.488 false reviews per true positive. Hybrid V1 generated the largest queue (23.3% flagged) and 1.822 false reviews per true positive. Rule-only validation had the smallest alert rate but captured only 2.3% of anomalies.

Table 6. Operational burden over 687,855 out-of-fold record–experiment observations.

Method	Flagged	Flag rate	True positives	False positives	Missed	Capture rate	False reviews/TP
AI consensus	118,799	17.3%	68,720	50,079	102,549	40.1%	0.729
Hybrid V3	103,554	15.1%	41,621	61,933	129,648	24.3%	1.488
Hybrid V1	159,956	23.3%	56,678	103,278	114,591	33.1%	1.822
Rule-only	42,579	6.2%	3,945	38,634	167,324	2.3%	9.793

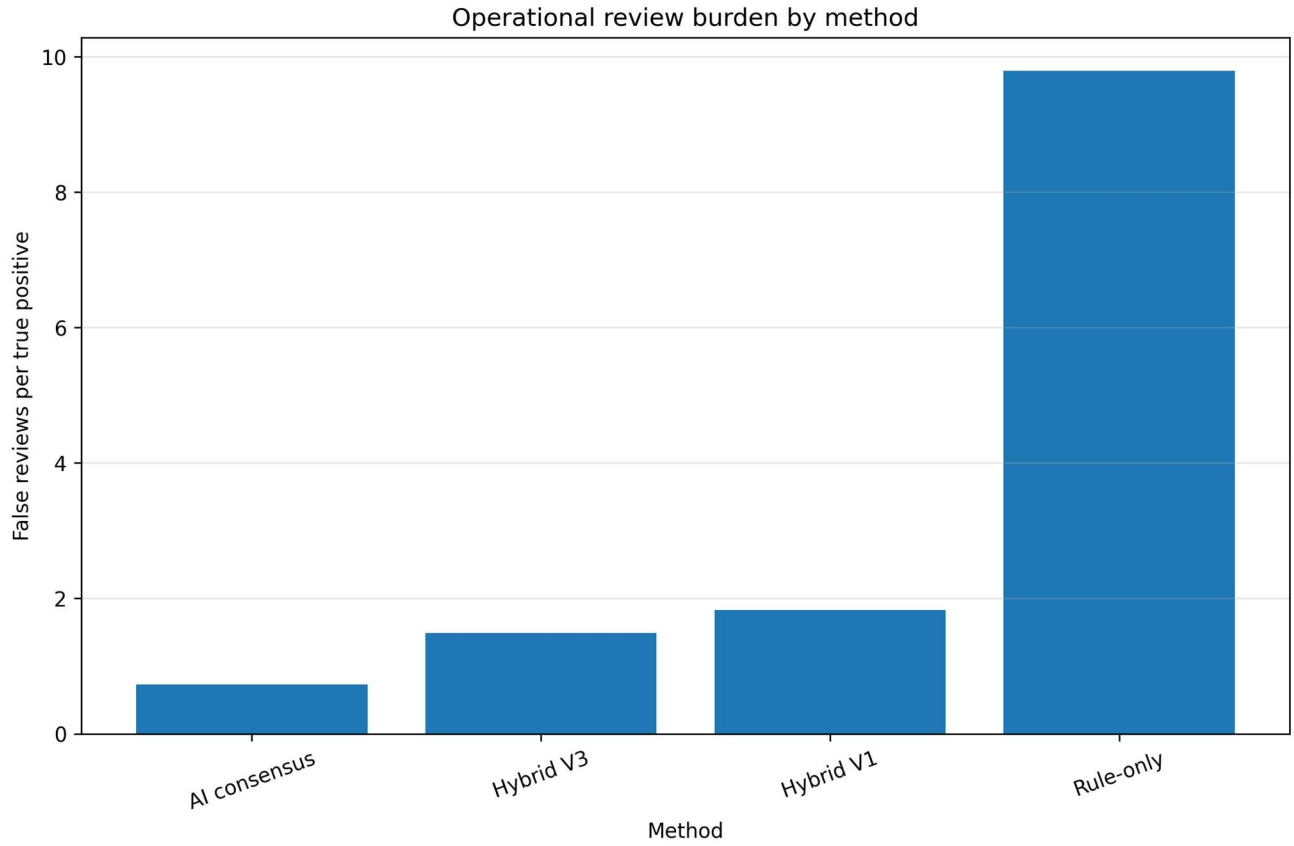


Figure 6. False-positive review burden per true-positive anomaly identified.

6. Discussion

6.1 Predictive accuracy and governance value are distinct

The main result is not that hybrid evidence fusion replaces the strongest AI baseline. AI consensus had the highest numerical macro-F1 and the best operational review efficiency. The F1 difference from Hybrid V3 was statistically inconclusive rather than evidence of equivalence or superiority. A defensible interpretation is that the two methods were not separated by the present experiment set at the corrected significance level, while their error profiles and operational burdens differed.

The value of Hybrid V3 lies in explicit evidence integration and governed action routing. It combines known business constraints with learned evidence, records uncertainty and conflict, and creates reproducible risk and trust scores. These properties are not captured by F1 alone. They support audit, policy versioning, and differentiated handling of known-correctable defects versus high-risk or ambiguous records.

The significant FPR reduction from Hybrid V1 demonstrates the importance of constrained calibration. An intuitively reasonable hybrid can perform worse operationally when rule and AI signals are combined without controlling alert burden. Nested experiment-level selection produced a more conservative, stable policy and avoided treating the initial hand-set weights as a validated design.

The fixed linear form should be interpreted as an inspectable governance artifact rather than as algorithmic novelty. Its advantage is that evidence contributions and intervention thresholds remain explicit; its cost is limited ability to model interactions among signals. A learned or probabilistic combiner could yield different predictive trade-offs. The present evidence therefore supports the policy form on auditability and stability grounds, not on a claim that deterministic fusion is superior to learned fusion.

6.2 Domain dependence and policy design

The finance and retail gains indicate that rule evidence can complement detector consensus for some combinations of schema constraints and anomaly types. The healthcare loss shows that a global policy can suppress recall when the domain contains different prevalence, density, or score distributions. High healthcare precision but low recall suggests that a domain-specific threshold or hierarchical calibration may be preferable where missed anomalies have a high cost.

These results imply that policy design should be risk-sensitive. A single global FPR tolerance treats false positives uniformly, but the cost of quarantining a retail invoice, escalating a financial record, or missing a healthcare utilization anomaly may differ substantially. Future work should replace one global objective with domain-specific cost functions and explicit service-level constraints.

6.3 Implications for data and knowledge engineering

The framework illustrates an interface between data engineering and knowledge engineering. Data-engineering components create reproducible features, scores, versions, and lineage. Knowledge-engineering components encode rule semantics, evidence interpretation, uncertainty handling, and action policy. Neither layer alone provides the complete governed decision.

For a lakehouse implementation, the evidence table can be materialized as a versioned transactional dataset. Each policy execution can write a decision record containing source-table version, rule-set version, model version, threshold, evidence, and action. This design enables replay after a policy change, comparison of decision distributions, and downstream enforcement through curated tables or quarantine zones.

The negative accuracy result is also useful for system design. It warns that combining more evidence sources does not automatically create a better detector. Hybridization should be justified by governance requirements and validated under explicit operational constraints rather than promoted as a guaranteed accuracy improvement.

6.4 Reproducibility and transparency

The experiment registry, fixed random seeds, injected ground truth, experiment-level splits, policy JSON files, record-level Parquet outputs, statistical result tables, and audit files form a reproducible evidence chain. Public source datasets and executable notebooks make it possible to rerun the study without access to proprietary enterprise data. Reporting non-significant and adverse results—particularly the higher Hybrid V3 FPR relative to AI consensus and weaker healthcare F1—reduces the risk of overstating the framework’s contribution.

7. Threats to validity and limitations

- Controlled anomalies. Injected defects provide precise ground truth but may not capture the full dependence, temporal evolution, or adversarial complexity of production incidents. The results establish behavior under a controlled taxonomy, not field effectiveness in every lakehouse.
- Limited detector family. The study uses two classical unsupervised methods. Deep reconstruction, probabilistic, graph, and streaming detectors could change the consensus and calibration landscape.
- Fusion comparator. The study does not compare the explicit weighted policy with logistic stacking, nonlinear meta-models, ensemble selection, or Bayesian model averaging. Such methods may improve predictive performance, but require separate controls for out-of-fold meta-training, calibration, retraining, drift, and explanation. The present results should therefore not be interpreted as evidence that policy-based linear fusion is the best available ensemble method.
- Global policy. Hybrid V3 uses one set of weights and one threshold across domains. The healthcare result indicates that domain-specific calibration and cost-sensitive objectives deserve separate evaluation.
- Operational action evaluation. Accept, repair, quarantine, and escalate actions are formally specified and logged, but the study does not include a human-review trial measuring reviewer agreement, time savings, or decision quality.

- Explanation boundary. The evidence vector and rule reasons are interpretable, but the study does not evaluate explanation fidelity or user comprehension with dedicated explanation metrics. It should not be read as a complete explainable-AI evaluation.
- Infrastructure generalization. The prototype uses public tabular data and cloud-oriented processing artifacts. Throughput, latency, and cost under sustained enterprise-scale streaming or multi-tenant workloads were outside the evaluated research questions.
- No formal equivalence claim. A non-significant F1 difference does not prove practical equivalence. A pre-specified non-inferiority margin and a larger number of independent datasets would be required for that conclusion.

8. Conclusion

This paper presented a governance-aware, policy-based framework for data-quality anomaly decisions. The framework combines deterministic rule risk, AI consensus, detector voting, and uncertainty into a bounded risk score and complementary trust index, then routes records to accept, repair, quarantine, or escalate actions while preserving the evidence and policy version required for audit and replay.

Across 63 experiments and three domains, AI consensus achieved the highest numerical macro-F1. Hybrid V3 was not significantly separated from AI consensus on F1 after Holm correction, but this non-significant result is not proof of equivalence or non-inferiority. Hybrid V3 had a significantly higher false-positive rate than AI consensus, significantly reduced false positives relative to the initial hybrid policy, significantly outperformed rule-only validation, and selected the same constrained configuration in all nested outer folds. Benefits remained domain-dependent: modest F1 gains appeared in finance and retail, while healthcare performance declined.

The contribution is therefore not a replacement for the strongest machine-learning baseline and not a claim of universal detection superiority. The study shows how knowledge-engineering principles can govern heterogeneous evidence through explicit, versioned, and replayable decision policies. The fixed policy improves inspectability but limits modeling flexibility. Future work should compare learned and probabilistic fusion, add modern deep anomaly detectors, evaluate domain-specific and cost-sensitive calibration, conduct human-review studies, and validate the action policy on production incidents.

Declarations

CRedit authorship contribution statement

Ramesh Babu Kallam: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing, Project administration.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The three source datasets are publicly available from the UCI Machine Learning Repository: Bank Marketing (<https://doi.org/10.24432/C5K306>), Diabetes 130-US Hospitals for Years 1999–2008 (<https://doi.org/10.24432/C5230J>), and Online Retail II (<https://doi.org/10.24432/C5CG6D>). Executable notebooks used to prepare, profile, inject anomalies, train detectors, and evaluate hybrid policies are supplied as supplementary material. Derived record-level outputs are large and are available from the corresponding author on reasonable request.

Ethics statement

The study used de-identified public datasets and did not recruit participants, intervene in care, or access private patient records. Institutional review board approval was therefore not required.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used OpenAI ChatGPT to assist with language editing, document structuring, and synthesis of verified code outputs into manuscript prose. The author reviewed and edited all generated material and takes full responsibility for the content of the article. Generative AI was not used to create or alter the source datasets, injected ground-truth labels, model outputs, or statistical results.

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